Block 1 — Week 4

Data Wrangling and Exploratory Analysis with R and Python

Essentials of Biomedical Informatics — Self-Paced Module (Asynchronous)

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Overview

This is the final week of Block 1. You'll take the cohort you built with SQL in Week 3 and put it to work: joining in a clinical measurement, characterizing the cohort, handling missing data, and visualizing a trend — side-by-side in R and Python. Plan for roughly 3–4 hours.

Companion Files

- synthetic_omop.duckdb — same database as Weeks 2–3
- Week4_Wrangling_R.qmd — worked tidyverse/ggplot2 walkthrough + exercise
- Week4_Wrangling_Python.ipynb — worked pandas/matplotlib walkthrough + exercise

You only need to complete ONE language track to fulfill this week's deliverable, but comparing both is a great way to see how the same logic reads in each language.

Learning Objectives

- Join a cohort definition to a measurement table and reshape the result for analysis
- Identify and appropriately handle missing values in a health dataset
- Build a simple "Table 1" cohort characterization table
- Create a clear, labeled visualization of a clinical trend over time

Core Concepts

R's tidyverse (dplyr, ggplot2) and Python's pandas/matplotlib solve the same problems with different syntax: dplyr's pipe (|>) chains verbs like filter(), mutate(), and summarise(), while pandas chains methods directly on a DataFrame (.query(), .assign(), .groupby()). Neither is objectively better — the goal of this course is that you can read and write both, and choose whichever your team or project already uses.

Missing data is not an edge case in health data — it is the default condition. Labs get ordered but never resulted, patients miss follow-up visits, fields go uncollected. The two mistakes to avoid are: silently dropping missing rows without checking how much data that removes, and imputing values without stating your assumption. The minimum standard, used throughout this course, is to always report a missingness rate before deciding how to handle it.

A "Table 1" is the standard first table in almost any clinical research paper: before showing any result, you characterize who is actually in the cohort (sample size, demographics, key baseline values). It exists so a reader can judge whether your cohort resembles the population they care about.

Finally, reproducible workflows: Quarto (.qmd) and Jupyter (.ipynb) both interleave code, output, and narrative text in one document that renders to HTML — this is what makes an analysis something a colleague (or you, months later) can re-run and verify, rather than a black box that produced a number.

Setup

Builds on Weeks 1–3. One R addition:

- R: `install.packages("lubridate")` (for simple date arithmetic; dplyr and ggplot2 are already installed from Week 1)
- Python: pandas and matplotlib (already installed in Week 1)

This Week's Deliverable

Work through Part 1 (Walkthrough) in EITHER Week4_Wrangling_R.qmd or Week4_Wrangling_Python.ipynb, then complete Part 2 (Your Turn) in that same file:

1. Rebuild the Hypertension + Lisinopril cohort from Week 3 and join it to systolic blood pressure measurements.
2. Report the missingness rate for those measurements.
3. Produce a Table 1 for this cohort.
4. Produce a spaghetti plot of blood pressure over time since starting lisinopril.
5. Write 2–3 sentences interpreting the plot and the missingness rate.

Submit the completed file (plus rendered .html for the R track) via the Week 4 dropbox in Brightspace.

Self-Check Quiz

Use this to check your own understanding before submitting. Answer the quiz in Brightspace.

1. Why should you report a missingness rate before dropping missing rows?

- a) It is a formal legal requirement
- b) Silently dropping data without checking how much you're removing can bias your results if the missingness isn't random
- c) It makes the code run faster
- d) It is not actually necessary

2. What is the purpose of a “Table 1” in clinical research?

- a) To list every variable in the raw dataset
- b) To characterize the cohort (demographics, baseline values) before presenting results about them
- c) To show only statistically significant findings
- d) To replace the need for a methods section

3. In the spaghetti plot exercise, why did we restrict the data to a specific window of days around treatment start?

- a) Because DuckDB cannot handle large date ranges
- b) Measurements from years before or after aren't relevant to this treatment's effect and would clutter the plot
- c) Because R and Python cannot plot more than 100 points
- d) It was not necessary; the window was arbitrary

4. Which of these is true about dplyr (R) and pandas (Python)?

- a) They solve completely different problems
- b) They solve the same data-wrangling problems with different syntax, and both are worth knowing
- c) pandas cannot join tables
- d) dplyr cannot filter rows

5. What makes a Quarto or Jupyter document “reproducible”?

- a) It uses a proprietary file format

- b) It interleaves code, output, and narrative so someone else can re-run it and verify the result
- c) It hides the underlying code from the reader
- d) It only works on the original author's computer

Suggested Readings

- Wickham & Grolemund, R for Data Science (2e) (free online: https://r4ds.hadley.nz/)
- Wes McKinney, Python for Data Analysis, 3E, (free online: https://wesmckinney.com/book/)

Suggested Grading Rubric

A pass grade required 60/100 points

Criterion	Weight	What to look for
Walkthrough completion	15%	All Part 1 cells/chunks run without errors
Cohort + missingness reported	25%	Correct join; missingness rate explicitly stated
Table 1	20%	Accurate summary values, clearly labeled
Visualization	20%	Appropriate chart type, labeled axes, sensible time window
Interpretation	20%	Shows genuine reasoning about the trend and missingness

What Now?

Across these four weeks, you move from installing tools to building a real cohort from a standardized clinical data model and analyzing it in two languages — a complete (if compact) reproducible research workflow. It will help you continue a journey into descriptive/inferential statistics and machine learning, building directly on the cohort-building and wrangling skills established here.